

FELIX MA

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Skills

Languages: C++, Python, Typescript, Javascript, C, Java, SystemVerilog, Verilog

Developer Tools: Git, Visual Studio, Jira, GitHub, BitBucket, GitLab, Quartus, ModelSim, FPGA

Education

University of British Columbia

September 2021 – April 2027

Bachelor of Applied Science in Computer Engineering

Vancouver, BC

Relevant Courses: Data Structures and Algorithms (C++), Operating Systems (C), Digital Systems Design (SystemVerilog, Quartus, FPGA Programming), Software Engineering (Java)

Work Experience

UBC Robotics and Control Laboratory

January 2026 - Present

Research Software Engineer Co-op

Vancouver, BC

- Organized **Python** research code into clear pipelines, with thought-out hierarchies of classes and inheritance, making the codebase more robust and modular
- Designed and implemented a comprehensive experiment-running framework, capable of running code analysis on a dataset using 50 different starting conditions; results were automatically collected, culminating in a research paper to be submitted to the International Ultrasonics Symposium 2026

Transoft Solutions

May 2025 – December 2025

Software Engineer Co-op

Burnaby, BC

- Wrote **C++** code making use of **Microsoft Foundation Class** dialogs which added comprehensive testing functionality, allowing users to automatically draw test cases for selected vehicles in **AutoCAD**, increasing possible test coverage from 10 vehicles to any subset in database of **2680+** based on chosen parameters
- Improved **Python** querying application by spot testing functions and simplifying algorithms, allowing users to query database of 2680+ vehicles over **10x** faster than previously possible
- Implemented new windows and elements in **Python** application using **PyQt** and **Jsonyx**, allowing users to view **JSON** files as a tree and easily make batch edits to common fields

Excelar

September 2024 – May 2025

Software Developer Co-op

Burnaby, BC

- Designed and implemented a responsive 'User Management' web page for proprietary application using **Angular 18**, allowing moderators to view all users in database in a single responsive table, improving the number of pages a user needed to navigate from **15+ to 1**
- Utilized **Bitbucket** and **Jira** for source control, managing development tasks, and facilitating code reviews
- Used **RxJS** functions in the creation of multi-page AI Package Deployment page with responsive status updates, allowing users to deploy medical AI models via the webpage instead of command line

Technical Projects

RC4 Encryption Breaker

June 2024

- Utilized built-in RAM and ROM on **FPGA** board for the purposes of storing and reading encrypted data
- Designed **5 finite state machines** consisting of **5-12 states** each with **SystemVerilog**, each working in tandem to crack encrypted messages via brute-force
- Verified correct operation of **FPGA** code by writing test benches and simulating all modules on **ModelSim**

FPGA Tone Organ

May 2024

- Developed a frequency divider module with SystemVerilog in order to split a clock signal into various audio frequencies
- Programmed **FPGA** and synthesized SystemVerilog code via **Quartus**, allowing the board to be used as a tone player
- Conducted thorough testing and verification of implementation with **SignalTap** and waveform simulations to ensure correct frequencies and switch behaviour

OS161 Operating System

September 2023 – December 2023

- Made use of **C** data structures and algorithms to write file management and low-level system calls in OS161, allowing the kernel to read/write files, efficiently manage file structures, and enhance overall system functionality
- Implemented core subsystems in OS161, including **synchronization**, **process signaling**, and **memory management**
- Leveraged **GDB** to perform kernel debugging using step-through and stack viewer to resolve faults in OS161